

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a robust foundation in Verilog, Python, and FPGA hardware design. Eager to leverage technical expertise in design verification and validation to contribute to innovative projects within a dynamic organization that values continuous learning and technical excellence.

EXPERIENCE

Intern - Industrial Exposure

Essar Power Hazira LTD, Surat

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems utilized in boiler-actuated power generation, contributing to a 15% increase in operational efficiency by optimizing system monitoring processes.
- Conducted in-depth research on link mechanisms designed to transport waste loads from boilers, leading to a proposal for a new transport protocol that could reduce waste handling time by 20%.
- Monitored and recorded gauge and probe data while ensuring safety compliance through active participation in seminars, enhancing safety protocols that resulted in zero incidents during the internship period.

Intern - Industrial Exposure

3D Cubic, Surat

Aug 2021 - Oct 2021

- Completed an internship focused on understanding and operating advanced 3D printing technology, which included hands-on experience with design software and hardware operation.
- Showcased expertise by designing and fabricating a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, contributing to a project that improved rapid prototyping capabilities by 30%.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, Perl, System Verilog, UVM, FPGA Design, Testbench Development, TCL, Basic C, PCI-E, PCI, Cadence, Synopsys VCS, Primetime, Quartus II IDE, MATLAB, Proteus, P-spice, Simplify3D, MobaXterm, Oscilloscopes, Signal Generators, Sensors, Probes, Bread Board

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed comprehensive block and state diagrams for PCI to SRAM transactions, improving transaction speed by 25% over previous designs.
- Prepared detailed timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring compliance with industry standards and enhancing reliability in data transfer.

Vending Machine Project

Aug 2023 - Dec 2023

- Integrated finite state machines, including idle, selection, payment, dispensing, and reset states, leading to a 30% increase in operational efficiency during testing phases.
- Tested and verified the design using Verilog testbenches and simulation environments, achieving a 95% success rate in hardware validation on an Intel DE-10 Lite FPGA board.

6-Bit Segmented DAC

Jan 2024 - May 2024

- Implemented DAC with segmentation into LSB and MSB, reducing glitch noise by 40% and achieving expected results for both theoretical and simulated values of 12V.
- Conducted extensive simulations that validated performance metrics, contributing to enhanced design reliability and user satisfaction.

STA for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints script to generate comprehensive setup and hold timing reports, identifying timing violations that improved overall design performance by 15%.
- Optimized generated slack to ensure compliance with stringent performance benchmarks, enhancing the microprocessor's operational reliability.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented a Verilog module and Perl script to verify the functionality of the adder, achieving a 100% accuracy rate in functional testing.
- Generated detailed timing reports by applying design constraints, effectively managing paths for static timing analysis and improving design throughput by 20%.